

Lab 2 Containerization & Kubernetes Orchestration

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1. Implement Kubernetes deployment- Implement Kubernetes deployment strategies such as rolling updates and rollbacks, and demonstrate scaling of the application replicas.

```
Last login: Tue Sep 15 09:39:38 on ttys000
anushkaberlia@Anushkas-MacBook-Air ~ % kubectl version --client
kubectl cluster-info
kubectl get nodes
Client Version: v1.36.1
Kustomize Version: v5.8.1
Kubernetes control plane is running at https://127.0.0.1:63487
CoreDNS is running at https://127.0.0.1:63487/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
NAME                STATUS    ROLES    AGE     VERSION
desktop-control-plane Ready    control-plane 108s    v1.36.1
anushkaberlia@Anushkas-MacBook-Air ~ % mkdir k8s-demo && cd k8s-demo
anushkaberlia@Anushkas-MacBook-Air k8s-demo % nano deployment.yaml
anushkaberlia@Anushkas-MacBook-Air k8s-demo % nano service.yaml
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
kubectl get pods -o wide
kubectl get svc
deployment.apps/my-app created
service/my-app-service created
NAME                READY    STATUS             RESTARTS   AGE   IP              NODE                NOMINATED NODE   READINESS GATES
my-app-c67564558f-2bl141 0/1      ContainerCreating   0           0s    <none>          desktop-control-plane <none>            <none>
my-app-c67564558f-2bzr1 0/1      ContainerCreating   0           0s    <none>          desktop-control-plane <none>            <none>
my-app-c67564558f-884p5 0/1      ContainerCreating   0           0s    <none>          desktop-control-plane <none>            <none>
NAME                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes           ClusterIP           10.96.0.1      <none>          443/TCP          18m
my-app-service       NodePort            10.96.19.125   <none>          80:30080/TCP     0s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl get pods
NAME                READY    STATUS             RESTARTS   AGE
my-app-c67564558f-2bl141 1/1      Running            0           48s
my-app-c67564558f-2bzr1 1/1      Running            0           48s
my-app-c67564558f-884p5 1/1      Running            0           48s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % curl http://localhost:30080
curl: (7) Failed to connect to localhost port 30080 after 6 ms: Couldn't connect to server
anushkaberlia@Anushkas-MacBook-Air k8s-demo % curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl scale deployment my-app --replicas=5
```

```
</html>
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl scale deployment my-app --replicas=5
kubectl get pods
deployment.apps/my-app scaled
NAME                READY    STATUS             RESTARTS   AGE
my-app-c67564558f-2bl141 1/1      Running            0           3m22s
my-app-c67564558f-2bzr1 1/1      Running            0           3m22s
my-app-c67564558f-4wtvr 0/1      ContainerCreating   0           0s
my-app-c67564558f-884p5 1/1      Running            0           3m22s
my-app-c67564558f-mncbm 0/1      ContainerCreating   0           0s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl scale deployment my-app --replicas=2
kubectl get pods
deployment.apps/my-app scaled
NAME                READY    STATUS             RESTARTS   AGE
my-app-c67564558f-2bl141 1/1      Running            0           3m27s
my-app-c67564558f-2bzr1 1/1      Running            0           3m27s
my-app-c67564558f-4wtvr 1/1      Terminating       0           5s
my-app-c67564558f-884p5 1/1      Terminating       0           3m27s
my-app-c67564558f-mncbm 1/1      Terminating       0           5s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl scale deployment my-app --replicas=3
kubectl get pods
deployment.apps/my-app scaled
NAME                READY    STATUS             RESTARTS   AGE
my-app-c67564558f-2bl141 1/1      Running            0           3m37s
my-app-c67564558f-2bzr1 1/1      Running            0           3m37s
my-app-c67564558f-lrjxj 0/1      ContainerCreating   0           0s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl set image deployment/my-app my-app-container=nginx:doesnotexist
kubectl rollout status deployment/my-app
deployment.apps/my-app image updated
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
deployment "my-app" successfully rolled out
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl rollout history deployment/my-app
deployment.apps/my-app
REVISION    CHANGE-CAUSE
1            <none>
2            <none>
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl set image deployment/my-app my-app-container=nginx:doesnotexist
deployment.apps/my-app image updated
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl get pods
NAME                READY    STATUS             RESTARTS   AGE
my-app-c6cd4b94567-6g8vh 1/1      Running            0           33m
my-app-c6cd4b94567-ppg4d 1/1      Running            0           34m
my-app-c66568b976-lrxjx 0/1      ErrImagePull       0           9s
my-app-c66568b976-mccr1 0/1      ErrImagePull       0           9s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl rollout undo deployment/my-app
kubectl rollout status deployment/my-app
Warning: resource deployments/my-app was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/my-app rolled back
```

```

my-app-6c7564558f-mnccm 1/1 Terminating 0 5s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl scale deployment my-app --replicas=3
kubectl get pods
deployment.apps/my-app scaled
NAME READY STATUS RESTARTS AGE
my-app-6c7564558f-2bl4l 1/1 Running 0 3m37s
my-app-6c7564558f-2bzrl 1/1 Running 0 3m37s
my-app-6c7564558f-3lmcj 0/1 ContainerCreating 0 0s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl set image deployment/my-app my-app-container=nginx:1.27
kubectl rollout status deployment/my-app
deployment.apps/my-app image updated
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
deployment "my-app" successfully rolled out
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl rollout history deployment/my-app
deployment.apps/my-app
REVISION CHANGE-CAUSE
1 <none>
2 <none>
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl set image deployment/my-app my-app-container=nginx:doesnotexist
deployment.apps/my-app image updated
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl get pods
NAME READY STATUS RESTARTS AGE
my-app-6cd4b94567-6g8vh 1/1 Running 0 33m
my-app-6cd4b94567-ppg4d 1/1 Running 0 34m
my-app-766568b976-lrxjx 0/1 ErrImagePull 0 9s
my-app-766568b976-mcdfl 0/1 ErrImagePull 0 9s
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl rollout undo deployment/my-app
kubectl rollout status deployment/my-app
Warning: resource deployments/my-app was previously managed with 'kubectl apply'. Rolling back will not update the kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/my-app rollout back
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
Waiting for deployment "my-app" rollout to finish: 2 of 3 updated replicas are available...
deployment "my-app" successfully rolled out
NAME READY STATUS RESTARTS AGE
my-app-6cd4b94567-6g8vh 1/1 Running 0 33m
my-app-6cd4b94567-97ht9 1/1 Running 0 1s
my-app-6cd4b94567-ppg4d 1/1 Running 0 34m
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl describe deployment my-app | grep Image
Image:
anushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl rollout history deployment/my-app
deployment.apps/my-app
REVISION CHANGE-CAUSE
1 <none>
3 <none>
4 <none>
anushkaberlia@Anushkas-MacBook-Air k8s-demo %

```

2. Configure continuous deployment-

Configure continuous deployment using Kubernetes by integrating the deployment with the CI/CD pipeline and automating manifest application.

```

Last login: Tue Sep 15 20:26:47 on ttys002
anushkaberlia@Anushkas-MacBook-Air ~ % git --version
git version 2.50.1 (Apple Git-155)
anushkaberlia@Anushkas-MacBook-Air ~ % git config --global user.name "Anushkaberlia28"
git config --global user.email "anushkaberlia@gmail.com"
anushkaberlia@Anushkas-MacBook-Air ~ % cd ~/k8s-demo
anushkaberlia@Anushkas-MacBook-Air k8s-demo % git init
Initialized empty Git repository in /Users/anushkaberlia/k8s-demo/.git/
anushkaberlia@Anushkas-MacBook-Air k8s-demo % cd ~/k8s-cicd-demo
cd: no such file or directory: /Users/anushkaberlia/k8s-cicd-demo
anushkaberlia@Anushkas-MacBook-Air k8s-demo % cd
anushkaberlia@Anushkas-MacBook-Air ~ % cd ~/k8s-cicd-demo
cd: no such file or directory: /Users/anushkaberlia/k8s-cicd-demo
anushkaberlia@Anushkas-MacBook-Air k8s-demo % cd ~/k8s-demo
anushkaberlia@Anushkas-MacBook-Air k8s-demo % git add deployment.yaml service.yaml
git commit -m "Initial k8s manifests"
[main (root-commit) 129e1ba] Initial k8s manifests
2 files changed, 37 insertions(+)
create mode 100644 deployment.yaml
create mode 100644 service.yaml
anushkaberlia@Anushkas-MacBook-Air k8s-demo % cd ~/k8s-demo
git branch -M main
git remote add origin https://github.com/Anushkaberlia28/k8s-cicd-demo.git
git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 10 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 681 bytes | 681.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Anushkaberlia28/k8s-cicd-demo.git
 * [new branch] main -> main
branch 'main' set up to track 'origin/main'.
anushkaberlia@Anushkas-MacBook-Air k8s-demo % uname -m
arm64
anushkaberlia@Anushkas-MacBook-Air k8s-demo % mkdir -p .github/workflows
anushkaberlia@Anushkas-MacBook-Air k8s-demo % nano .github/workflows/cd.yml
anushkaberlia@Anushkas-MacBook-Air k8s-demo % cat .github/workflows/cd.yml
name: Kubernetes Continuous Deployment

on:
  push:
    branches:
      - main

jobs:
  deploy:
    runs-on: self-hosted

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Check Kubernetes
        run: kubectl cluster-info

      - name: Apply Kubernetes manifests
        run: |
          kubectl apply -f deployment.yaml
          kubectl apply -f service.yaml

```

```

on:
  push:
    branches:
      - main

jobs:
  deploy:
    runs-on: self-hosted

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Check Kubernetes
        run: kubectl cluster-info

      - name: Apply Kubernetes manifests
        run: |
          kubectl apply -f deployment.yaml
          kubectl apply -f service.yaml

      - name: Check deployment
        run: |
          kubectl rollout status deployment/my-app
          kubectl get pods
          kubectl get services

ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git add .github/workflows/cd.yml
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   .github/workflows/cd.yml

ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git commit -m "Add Kubernetes continuous deployment workflow"
[main 7f3d223] Add Kubernetes continuous deployment workflow
 1 file changed, 28 insertions(+)
 create mode 100644 .github/workflows/cd.yml
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 669 bytes | 669.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Anushkaberlia28/k8s-cicd-demo.git
 129e1ba..7f3d223  main -> main
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % cd ~/k8s-demo
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl get pods

```

```

steps:
  - name: Checkout code
    uses: actions/checkout@v4

  - name: Check Kubernetes
    run: kubectl cluster-info

  - name: Apply Kubernetes manifests
    run: |
      kubectl apply -f deployment.yaml
      kubectl apply -f service.yaml

  - name: Check deployment
    run: |
      kubectl rollout status deployment/my-app
      kubectl get pods
      kubectl get services

ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git add .github/workflows/cd.yml
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   .github/workflows/cd.yml

ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git commit -m "Add Kubernetes continuous deployment workflow"
[main 7f3d223] Add Kubernetes continuous deployment workflow
 1 file changed, 28 insertions(+)
 create mode 100644 .github/workflows/cd.yml
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 669 bytes | 669.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Anushkaberlia28/k8s-cicd-demo.git
 129e1ba..7f3d223  main -> main
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % cd ~/k8s-demo
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl get pods

```

```

NAME          READY   UP-TO-DATE   AVAILABLE   AGE
my-app        3/3     3             3            3h39m

ianushkaberlia@Anushkas-MacBook-Air k8s-demo % kubectl get services

```

```

NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes    ClusterIP   10.96.0.1     <none>        443/TCP    3h57m
my-app-service NodePort    10.96.19.125  <none>        80:30080/TCP 3h39m
ianushkaberlia@Anushkas-MacBook-Air k8s-demo % █

```

GITHUB SETUP

```
Last login: Tue Sep 15 21:16:06 on ttys001
anushkaberlia@Anushkas-MacBook-Air ~ % mkdir actions-runner && cd actions-runner
curl -o actions-runner-osx-arm64-2.337.0.tar.gz -L https://github.com/actions/runner/releases/download/v2.337.0/actions-runner-osx-arm64-2.337.0.tar.gz
tar xzf ./actions-runner-osx-arm64-2.337.0.tar.gz
./config.sh --url https://github.com/Anushkaberlia28/k8s-cicd-demo --token BCCAFSA4XF524G5DLNEDMCCCLKVF25I
mkdir: actions-runner: File exists
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0 --:--:--  0:00:01 --:--:--   0
59 121M   59 71.9M    0     0 32410      0  1:05:41  0:38:47  0:26:54   0
curl: (56) Recv failure: Connection reset by peer
./externals/node28/bin/node: truncated gzip input: Unknown error: -1
tar: Error exit delayed from previous errors.
./config.sh: line 88: ./bin/Runner.Listener: No such file or directory
anushkaberlia@Anushkas-MacBook-Air ~ % ./run.sh
/Users/anushkaberlia/run-helper.sh: line 36: /Users/anushkaberlia/bin/Runner.Listener: No such file or directory
Exiting with unknown error code: 127
Exiting runner...
anushkaberlia@Anushkas-MacBook-Air ~ % cd ~
rm -rf actions-runner
mkdir actions-runner
cd actions-runner
mv ~/Downloads/actions-runner-osx-arm64-2.337.0.tar.gz .
tar xzf actions-runner-osx-arm64-2.337.0.tar.gz
anushkaberlia@Anushkas-MacBook-Air actions-runner % ./config.sh --url https://github.com/Anushkaberlia28/k8s-cicd-demo --token BCCAFSA4XF524G5DLNEDMCCCLKVF25I
./config.sh: line 81: 18582 Killed: 9                  ./bin/Runner.Listener configure "$@"
anushkaberlia@Anushkas-MacBook-Air actions-runner % ./run.sh
/Users/anushkaberlia/actions-runner/run-helper.sh: line 36: 18667 Killed: 9                  *$DIR*/bin/Runner.Listener run $*
Exiting with unknown error code: 137
Exiting runner...
anushkaberlia@Anushkas-MacBook-Air actions-runner % ./run.sh./run.sh
zsh: no such file or directory: ./run.sh./run.sh
anushkaberlia@Anushkas-MacBook-Air actions-runner % xattr ./bin/Runner.Listener
com.apple.quarantine
anushkaberlia@Anushkas-MacBook-Air actions-runner % xattr -dr com.apple.quarantine ~/actions-runner
anushkaberlia@Anushkas-MacBook-Air actions-runner % cd ~/actions-runner
./config.sh --url https://github.com/Anushkaberlia28/k8s-cicd-demo --token BCCAFSA4XF524G5DLNEDMCCCLKVF25I
```

GitHub Actions

Self-hosted runner registration

```
# Authentication

Http response code: Not Found from 'POST https://api.github.com/actions/runner-registration' (Request Id: FB82:32EE5:F07EF:1031FC:6AA97B92)
{"message":"Not Found","documentation_url":"https://docs.github.com/rest","status":"404"}
Response status code does not indicate success: 404 (Not Found).
anushkaberlia@Anushkas-MacBook-Air actions-runner % CD
anushkaberlia@Anushkas-MacBook-Air actions-runner % cd
anushkaberlia@Anushkas-MacBook-Air ~ % cd ~/actions-runner
anushkaberlia@Anushkas-MacBook-Air actions-runner % ./config.sh --url https://github.com/Anushkaberlia28/k8s-cicd-demo --token YOUR_FRESH_TOKEN
```

GitHub Actions

Self-hosted runner registration

```
# Authentication

Http response code: Not Found from 'POST https://api.github.com/actions/runner-registration' (Request Id: FBDF:28CB7C:110CE4:134EA3:6AA97EFC)
{"message":"Not Found","documentation_url":"https://docs.github.com/rest","status":"404"}
Response status code does not indicate success: 404 (Not Found).
anushkaberlia@Anushkas-MacBook-Air actions-runner % ./config.sh --url https://github.com/Anushkaberlia28/k8s-cicd-demo --token BCCAFS2QLSOYAAQO7HMR3TKVGGA6
```

GitHub Actions

Self-hosted runner registration

```
# Authentication

✓ Connected to GitHub

# Runner Registration

Enter the name of the runner group to add this runner to: [press Enter for Default]

Enter the name of runner: [press Enter for Anushkas-MacBook-Air]

This runner will have the following labels: 'self-hosted', 'macOS', 'ARM64'
Enter any additional labels (ex. label-1,label-2): [press Enter to skip]

✓ Runner successfully added

# Runner settings

Enter name of work folder: [press Enter for _work]

✓ Settings Saved.

anushkaberlia@Anushkas-MacBook-Air actions-runner % ./run.sh

✓ Connected to GitHub

Current runner version: '2.337.0'
2026-09-15 17:26:11Z: Listening for Jobs
2026-09-15 17:33:19Z: Running Job: deploy
2026-09-15 17:33:52Z: Job deploy completed with result: Succeeded
```

